

Project 3: Glider Design And Performance

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Task 1: Glider Design

Figure 1 below shows a glider design created in OpenVSP that attempts to maximize range and endurance in given conditions. First, I wanted to maximize the aspect ratio to reduce induced drag caused by effects at wingtip. This was done by using the maximum allowed length given by the project rules, 7 meters. I also attempted to make the glider balanced by visually estimating wing placements that appeared to center the center-of-mass. Additionally, I made each wing section have a gradually increasing sweep to create better balance between the weight and the aerodynamic center, without sacrificing too much efficiency. Each section also has a twist which changes its angle of attack. Different angles of attack for each section will make it so that other sections can still produce lift when a certain angle causes a section to stall. Additionally, there are differing cambers, with the largest section, closest to the fuselage, having the greatest camber. The higher camber will increase the lift generated while minimizing drag. Because lift further away from the fuselage adds a weight penalty, I reduced their camber and also their span the further away they were from the fuselage. I also made the airfoils very thin to reduce overall weight and drag.

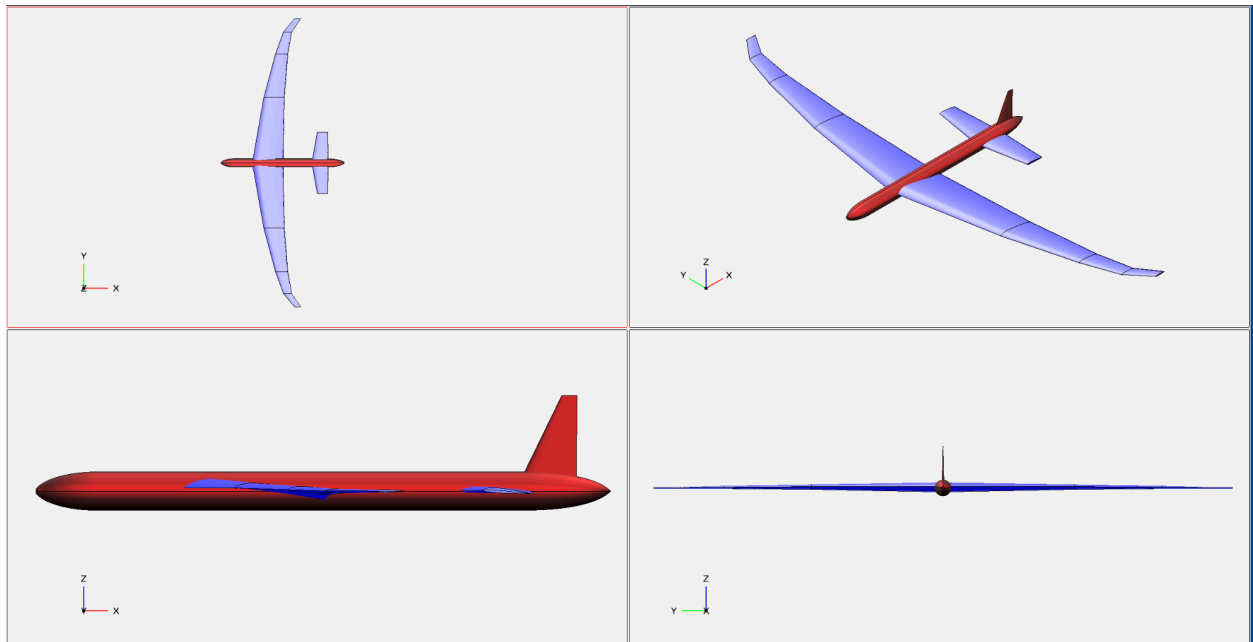


Figure 1: Top, Side, Front, and Isometric view of glider in OpenVSP

Task 2: Airfoil Section Profile

*note: I used a NACA 6210 in Xfoil as the NACA 6205 had many problems converging.

Below are the performance curves for the root and tip foils, NACA 6210 and NACA 0010, respectively, over the range of 0 to 15 degrees angle of attack.

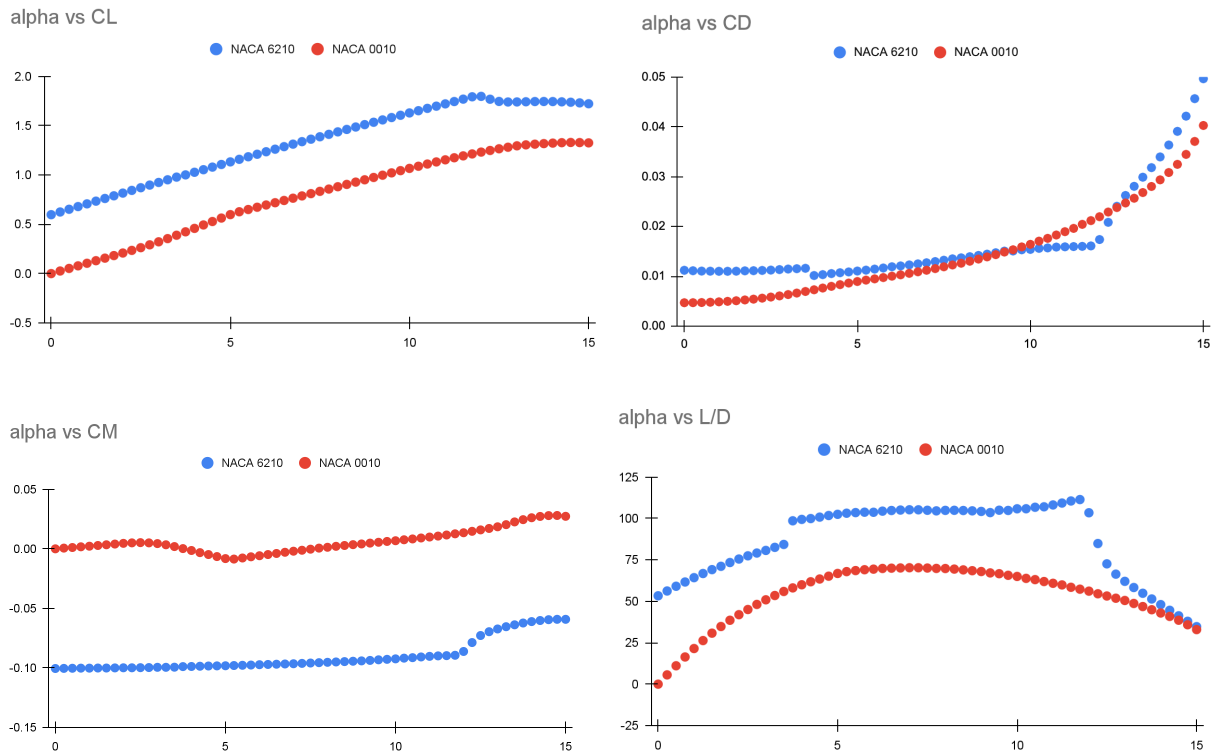


Figure 2: Alpha vs C_L (top left), C_D (top right), C_M (bottom left), and L/D (bottom right) for NACA 6210 and NACA 0010 foils from 0 to 15 degrees angle of attack

The NACA 6210 airfoil, with its camber, generates positive lift at zero angle of attack and continues to increase more significantly as the angle of attack increases compared to the symmetrical NACA 0010. Both airfoils show increasing lift coefficients with increasing angle of attack up to a point, after which the lift coefficient plateaus at around 12 degrees. Initially, both airfoils exhibit low drag, but as the angle of attack increases, the drag for both also increases. The increase in drag is relatively gradual for lower angles of attack and becomes more dramatic past a certain angle, indicating approaching separation. The NACA 6210 shows a slightly higher drag coefficient than the NACA 0010 for most of the angle of attack range. The moment coefficient for the NACA 6210 is negative and increases (becomes less negative) with the angle of attack, indicating a nose-up pitching moment which stabilizes as the angle of attack increases. The NACA 0010 has a moment coefficient that is nearly zero and slightly positive, indicating very little pitching moment. Both airfoils show an increase in L/D with increasing angle of attack after reaching a maximum. The NACA 6210 has a higher L/D ratio than the NACA 0010 for most angles of attack, indicating better efficiency. It also maintains its efficiency at a higher angle of attack at around 12 degrees, before plummeting down.

According to the graphs, the NACA 6210 seems to have a separation of about 12 degrees. It is a little harder to tell with the NACA 0010 because there are no dramatic increases or decreases in any of the graphs, but I would estimate the separation angle to be about 8 degrees.

The angle of maximum L/D for both foils occurred at the aforementioned, respective separation angles. Moments can negatively affect glider performance because pitching up too high could cause stall and pitching down will cause the glider to fall faster, while no pitch will maximize range for a given endurance. These performance graphs overall lead to better information about the efficiency of the airfoils. The NACA 6210 overall has high L/D over a large range of angles while negatively affect moment; and the NACA 0010 has overall worse performance but keeps a neutral moment.

Task 3: Flowfield and Body Force Analysis

Table 1 below shows the force and moments on the glider. The lift, drag, and moment were found on Simscale by adding the pressure and viscous contributions for each direction. The mass and center of gravity were calculated using OpenVSP, and the new mass and new center of gravity were hand-calculated.

Table 1. The forces and moments on the glider

F_z (lift)	804.142 N
F_x (drag)	44.929 N
M_y (spanwise moment)	-945.709 N-m
Mass of wing set	58.646 kg
Center of Gravity of wingset	1.395 m
New (total) Mass	118.646 kg
New Center of Gravity	

The following calculation was done to find the center of gravity and the center of pressure.

$$x_{CG} = \frac{\sum mx_m}{\sum m} = \frac{(58.484 \text{ kg})(1.395 \text{ m}) + (60 \text{ kg})(1 \text{ m})}{118.646 \text{ kg}} = 1.193 \text{ m}$$

$$x_{CP} = \frac{M_y}{F_z} = \frac{-945.709 \text{ Nm}}{804.142} = -1.176 \text{ m}$$

The center of pressure is 0.017 m closer to the nose of the glider than the center of gravity. Unfortunately, it is more ideal for the center of pressure to be behind the center of gravity because it provides flight stability. Because my center of pressure is ahead of the center of gravity, a slight change in pitch will cause the glider to pitch further from level flight. In contrast, if my center of pressure were to be behind the center of gravity, there would be stable

flight as the pressure would self-correct any change in pitch. Luckily, it is not that far away; therefore, it may take longer for the aircraft to go completely unstable.

I tried to depict as many aerodynamic concepts in one flow visualization. First, it is clear that there is low pressure on the top of the wing and high pressure on the bottom of the wing. There are many effects that cause this pressure differential, such as the curvature confining the flow to a smaller area, causing it to accelerate due to Bernoulli's principle depicted by the velocity flow tracers placed at the root. I also used the flow tracers to show how the Coanda effect causes the flow to stick to the surface of the wing and connect back at the trailing edge due to the Kutta condition. Circulation is required to satisfy this condition, which is a sign of lift. Finally, in red, I made another set of flow tracers to depict the vortices created by the wingtip, which creates induced drag.

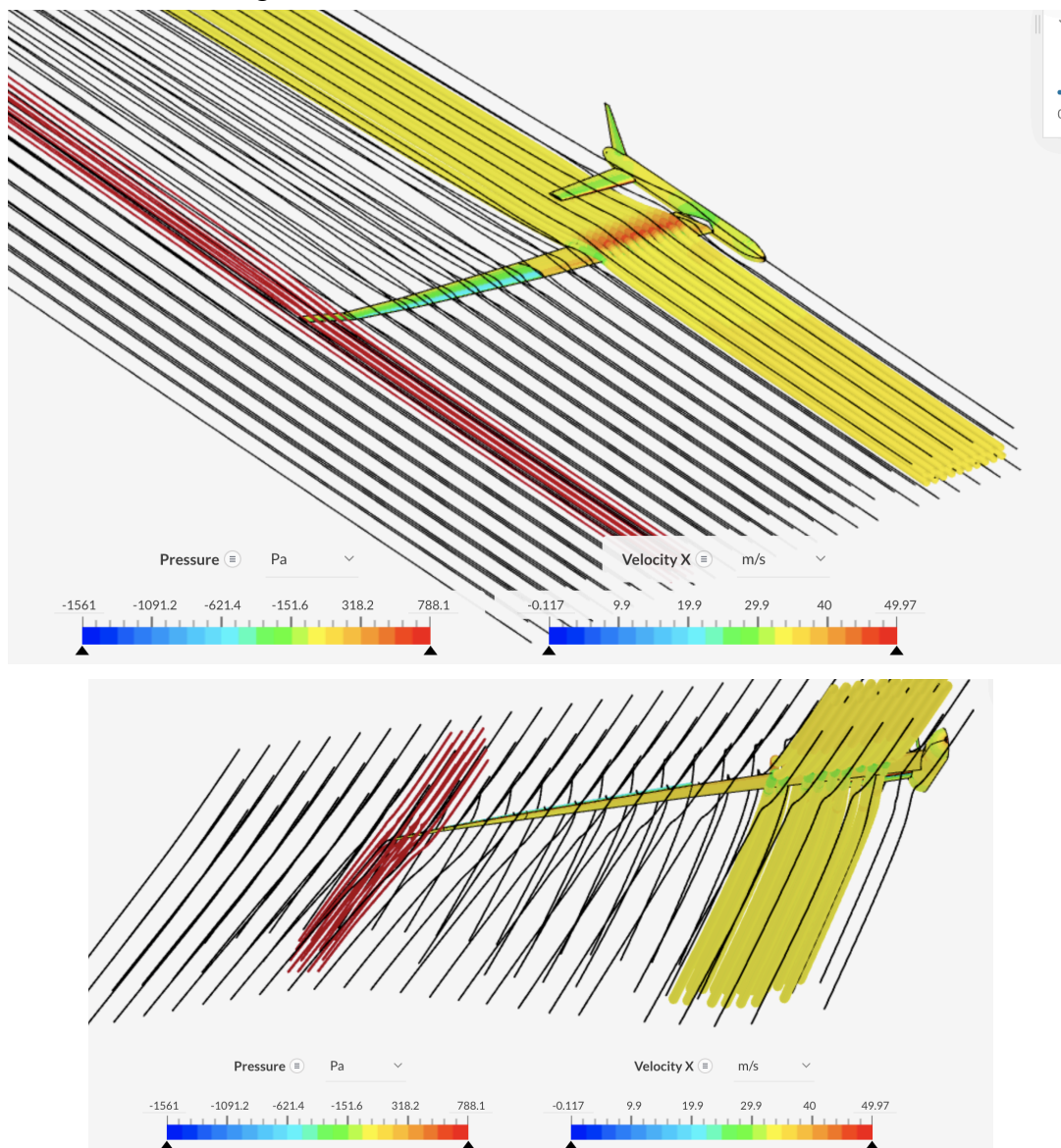


Figure 3: 3D Flow visualization on Simscale